

MAE 1001



Introduction to Mechanical and Aerospace Engineering

Course website: <https://gwu-mae1001.github.io/fall2025/>

Prof. Kartik Bulusu, CS Dept.

Graduate Teaching Assistant:

Preethi Siva Kumar (MAE Dept., MS student)

Learning Assistants:

Nathan Janssen (Senior, MAE Dept.)

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Shota Kakiuchi (Senior, MAE Dept.)

Tech support:

Rithik Gunuganti (CS Dept., MS student)



School of Engineering
& Applied Science

THE GEORGE WASHINGTON UNIVERSITY

Fall 2025

Photo: Kartik Bulusu

Seating Arrangement

Table 1 (8 students):

- Yahya Ahmed Ahmed Sadek Elsewedy
- Paige Romanowski
- Owen Fritts
- Omar Mohammed A I Al Emadi
- Lucas Narvaez
- Andy Pham
- Grace Ephrem
- Thomas Melotto

Table 2 (8 students):

- Holt Perryman
- Nabek Ababiya
- Novah Wali
- Kim Zamora
- Teresa Cherian
- Johnny Lopez Bertrand
- Emre Ozkazanc
- Shu Shu Wu

Table 3 (8 students):

- Irene Oh
- Addison Dooley
- Sena Mute
- Evan Agapiou
- Otto Erhart
- Stella Iyengar
- Nate Behabtu
- Henry Steffen

Table 4 (8 students):

- Marthine Jean Pierre
- Aidan Willett
- Chance Hwang
- Dennis Ventura
- Brandon Alvarado
- Camryn Lorencovitz
- Albert Wu
- Gael Merida Chavez

Table 5 (8 students):

- Jessica Gil
- Jocelyn Carrillo
- Tre Troy
- Malik Ali
- Tim Harrington
- Cody Keel
- Curtis Charlery
- Ryan Barks

Table 6 (7 students):

- Hamza Chaudry
- Daris Dedic
- Alex Lagrant
- Naomi Kim
- Dylan Nichols
- Cadence Cassidy
- Lilia Hammadi

Table 7 (7 students):

- Luz De La Cruz
- Wassime Ait-Si
- Ali Ahmed Y A Al Malki
- Alexander Lele
- Akina Pallera
- Mohammed Jabir H A Al Jabir
- Zaynah Asfour

Table 8 (7 students):

- Maeve Carey
- Cesar Tovar
- Zachary Bogan
- Saihaj Kaur
- Ethan Peralta
- Harper Barrios
- Evan Chukwunonye

Table 9 (7 students):

- Kazi Afra Saiara
- Daniel Zhang
- Eli Ecret
- Inti Stephenson
- Diego Flores
- Ivan De La Cruz
- Matheo Valdivia



Road Map for Class 3

30-Minute Session Blocks Breakdown

Photo Credits: Kartik Bulusu
Icon source: canva.com



**Blitz Talk with
Grace Welsh**



**Camera
Disassembly Lab**

Dissect a disposable lab



Selfie Lab

Camera and Face
Detection

**Introduction to
Week 3**

Announcements,
overviews and more!



**Guest Lecture with
Dr Saniya LeBlanc**

Energy conversion
technologies involving
advanced materials

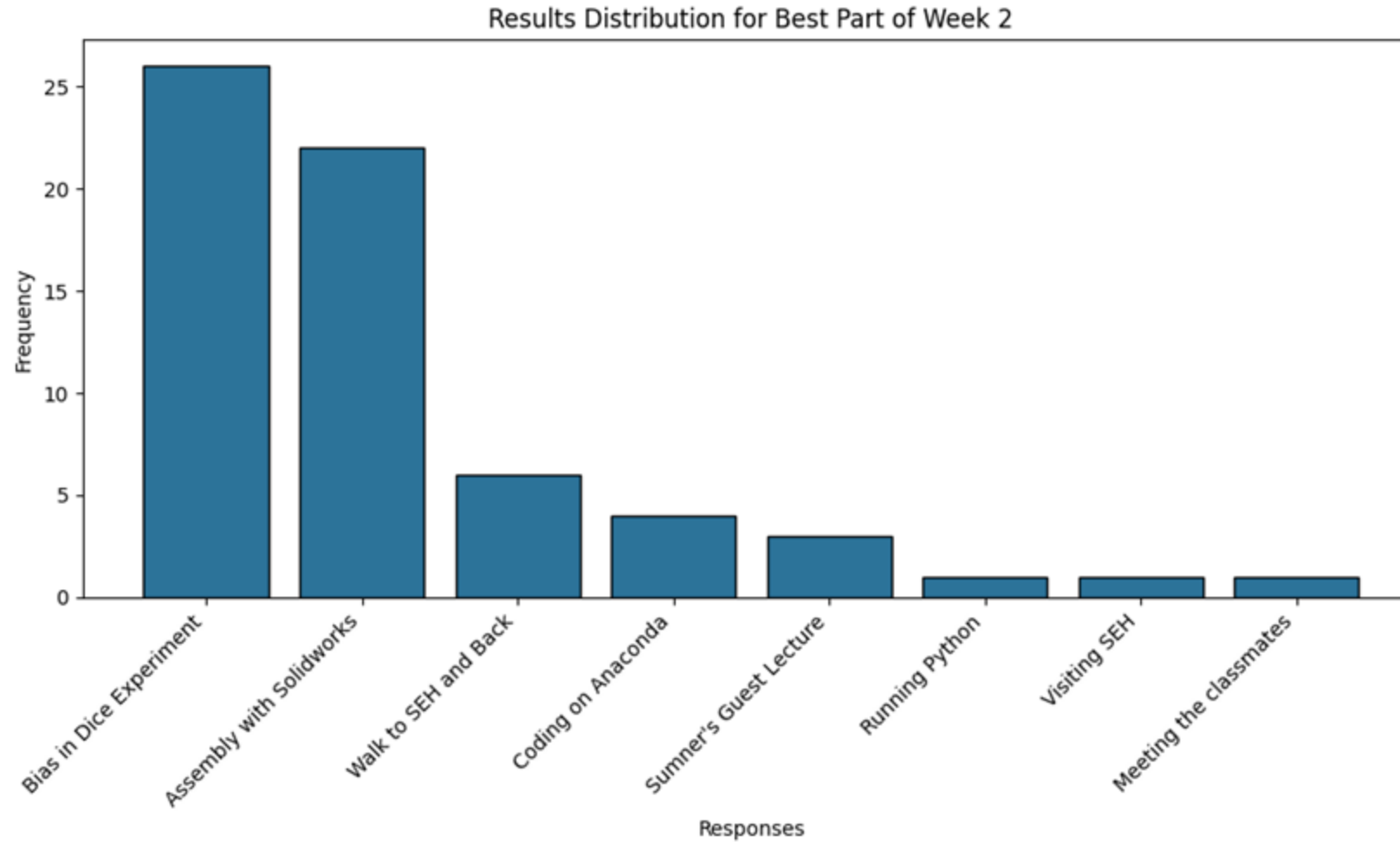


Python 101

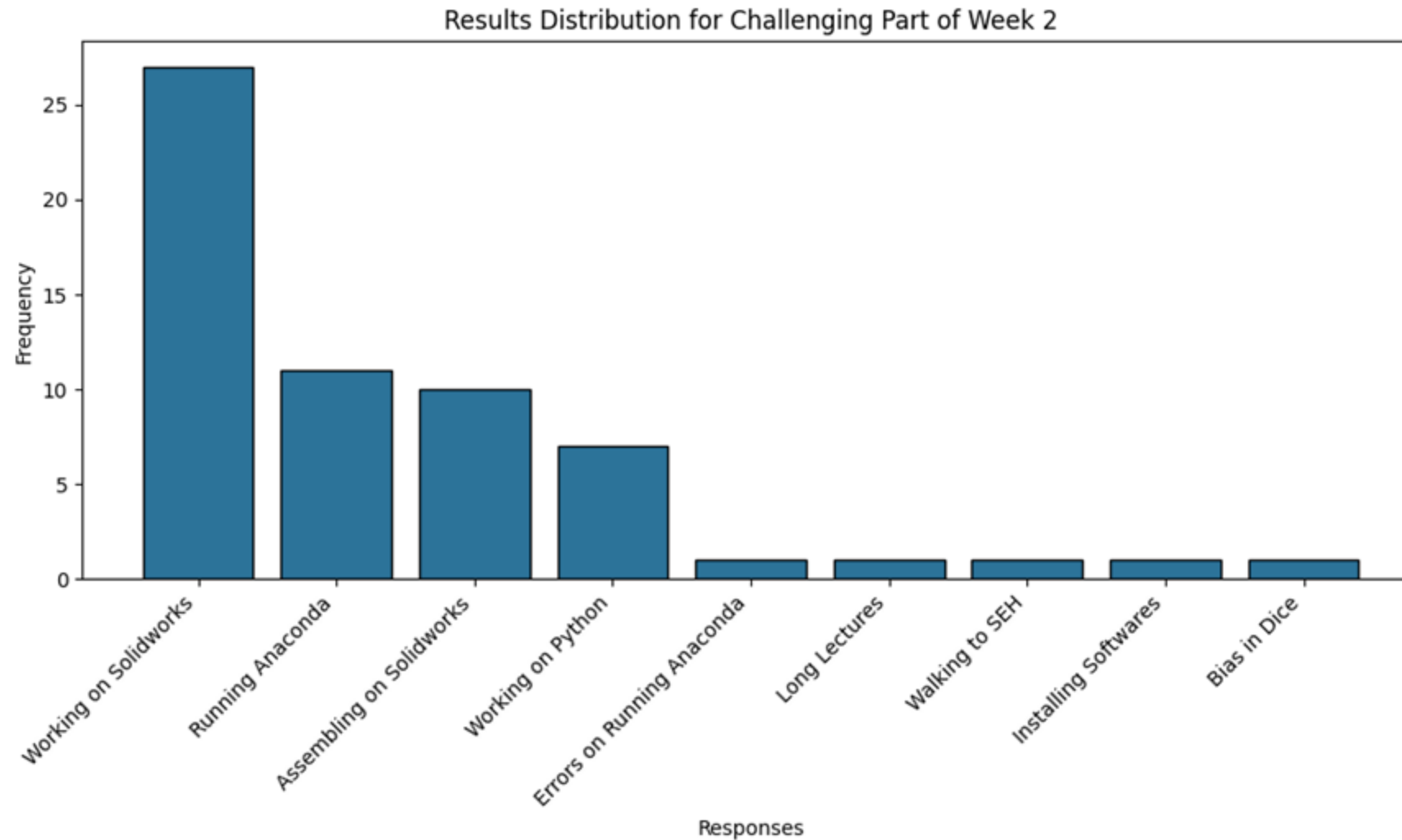
Learn the basics of
Python with Variables,
Functions and Loops



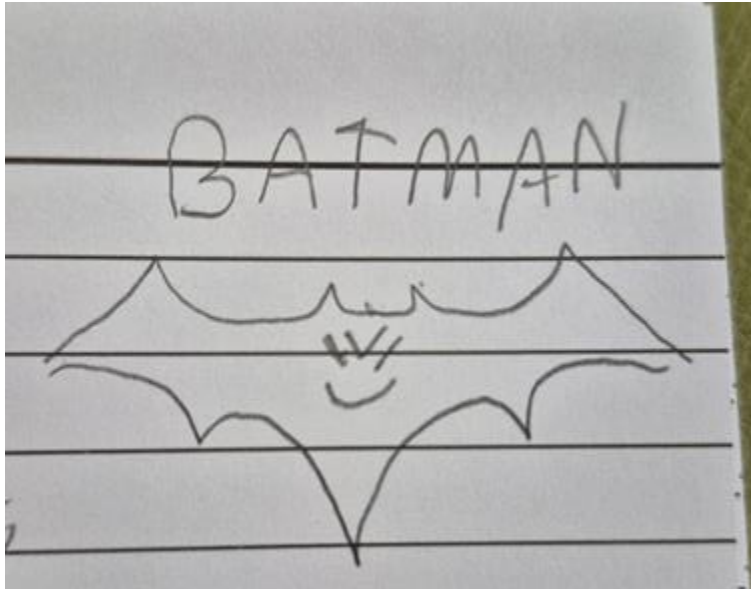
Week 2 - Best Part of the Class



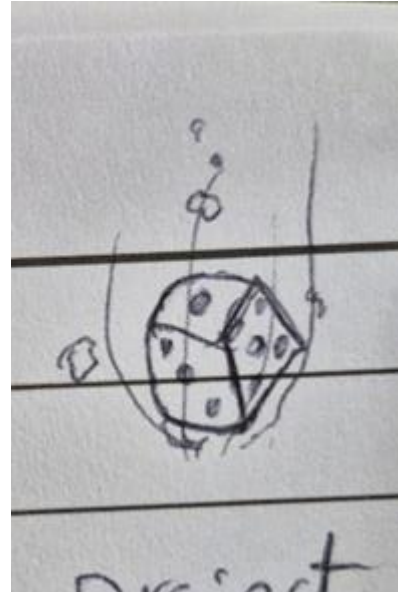
Week 2 - Challenging Part of the Class



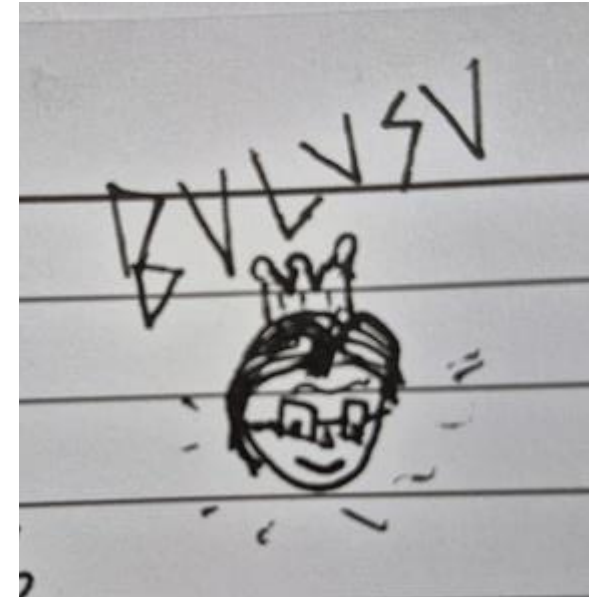
Week 2 - Best Doodles



Inti



Malik Ali



Cesar Tovar



Blitz Talk with Grace Welsh



Senior in Mechanical and Aerospace Engineering

Email: gracewelsh27@gwmail.gwu.edu

LinkedIn: [linkedin.com/in/grace-welsh-384b7326a/](https://www.linkedin.com/in/grace-welsh-384b7326a/)

Grace is a senior studying MAE and a Presidential Scholar at GWU. She has worked in composite material testing, design and manufacturing. She is a part of the rocket team (aerostructures), SEASSPAN, A.O.E. (Alpha Omega Epsilon), and a student research assistant with Professor Panah.



Guest Lecture with Professor Saniya LeBlanc



PhD Student working in the LeBlanc Lab

Email: sleblanc@gwu.edu

Website: leblanclab.com

Professor Saniya LeBlanc's research lies at the intersection of materials science, energy conversion, and thermal transport. Her group enables nanomaterials integration by bridging the divide between rapid, inexpensive manufacturing of nanostructures and device integration of nanoengineered components. Her experimental research includes development of scalable nanomanufacturing processes such as spray coating and printing of functional nanomaterials.

Projects involve energy harvesting devices that utilize the unique combination of properties offered by nanostructured materials; recent examples include thermoelectric power generators for waste-heat recovery. Combining energy system, cost, and policy analyses, she creates feasibility assessments for the scaling of nanotechnologies for energy applications.



Exit Ticket

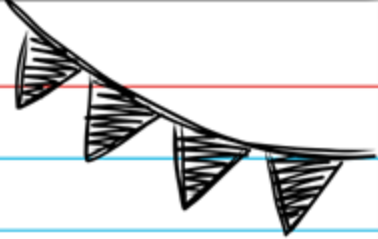
Name: _____

Three things you learnt today:

1. _____
2. _____
3. _____

Best Part of the Class: _____

Most Challenging Part of the Class: _____



Assignments for Week 3 - Due 11/13

In-Class Lab 3A: Bias in Dice
Experiment - Trial 2

In-Class Lab 3B: Camera Dissection
Lab

In-Class Lab 3C: Raspberry Pi Selfie
Upload

Assignment 3D: Complete Week 1
HW on Jupyter Hub ([bulusu-
py.ead.gwu.edu](https://bulusu-py.ead.gwu.edu))

